Intrinsic Generative Selectors: Finite Control, Arithmetic State, and Observational Realization

UNNS-U2-001 – version 1.0 (v001)

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Abstract

Let G be the graph on the positive integers in which kN is joined to $k(N+1)^2$ for all $k, N \geq 1$. A *selector* chooses one incident edge at every step, from the current integer and a finite control state only, with no external input. We prove three results. (i) No deterministic rule that reads only the current integer can pass through $1, 2, 3, \ldots$ in this order. (ii) A deterministic rule with 43 control states that reads the current integer by exact arithmetic *can*: it visits $1, 2, 3, \ldots$ in order, reaching each $n+1$ from n in at most six moves, and the unbounded information needed at each step is carried by the integer itself. (iii) The doubling distance is uniformly bounded, $\sup_n \mathrm{d}(n, 2n) = 5$. The construction in (ii) rests on nine path identities for $n \to n+1$ whose intermediate states are strictly increasing polynomials in a class parameter, which makes that parameter recoverable from the current state alone. We also give a precise definition of the selector classes and show why an unrestricted observation map would make any comparison between them empty.

1 Introduction

Provenance. The graph G is the undirected closure of the SELECT-II transformation $m \mapsto m(N+1)^2/N$ $(N \mid m)$ of the UNNS Substrate. Its connectivity and the uniform bound $\mathrm{d}(n, n+1) \leq 6$ are established in the preceding releases [3, 2]; the forward-only baseline is research note U0-001 [1]. The present release, UNNS-U2-001, leaves all of them unchanged, as it does the frozen reference collection [4, 5].

Question. Bounded distance says that every integer can be reached from its predecessor by a short path. It does not say that a fixed rule can *choose* that path. Can the arithmetic itself determine the next move, using a fixed rule and no instructions supplied by the desired output sequence?

Results.

Theorem 1.1 (Summary)**.** *(a)* (U2.0, memoryless obstruction.) *For every function $F : \mathbb{N}_{>0} \to \mathbb{N}_{>0}$ choosing a neighbour of its argument, no orbit $x_{j+1} = F(x_j)$ contains $1, 2, 3, \ldots$ as a subsequence.*

(b) (U2.1, finite-control realization.) *There is a deterministic selector Σ with $|Q| = 43$ control states, using exact integer arithmetic on the current state and no other input, whose trajectory from $x = 1$ returns to the ready control state exactly at $2, 3, 4, \ldots$ in this order, with at most six moves between consecutive ready states.*

(c) (Doubling.) $\mathrm{d}(n, 2n) \leq 5$ *for all n, with equality at $n = 1$.*

2 Graph facts used

Throughout, G has an edge $\{kN,\, k(N+1)^2\}$ for each pair $k, N \geq 1$; a move $+N$ from x requires $N \mid x$ and gives $x(N+1)^2/N$, and a move $-N$ requires $(N+1)^2 \mid x$ and gives $xN/(N+1)^2$. The moves $+N$ and $-N$ are mutually inverse.

Lemma 2.1 (Consecutive integers are never adjacent)**.** *For $n \geq 1$, $\{n, n+1\}$ is not an edge; hence $\mathrm{d}(n, n+1) \geq 2$.*

Proof. The larger endpoint of an edge is $(N+1)^2/N = N + 2 + 1/N > 4$ times the smaller, whereas $(n+1)/n \leq 2$. □

Lemma 2.2 (Nine path identities [3])**.** *For every $n \geq 1$ there is a path $n = x_0, x_1, \ldots, x_\ell = n+1$ with $\ell \leq 6$ and every step certified, given by one of the nine parametric families of Appendix A (for $n \in \{1, 3\}$, by an explicit path).*

Lemma 2.3 (Doubling)**.** $\sup_{n \geq 1} \mathrm{d}(n, 2n) = 5$.

Proof. If $\{x, y\}$ is an edge then so is $\{nx, ny\}$ (multiply k by n), so $\mathrm{d}(na, nb) \leq \mathrm{d}(a, b)$. The path $1 \to 4 \to 18 \to 81 \to 8 \to 2$ (moves $+1, +2, +2, -8, -1$) gives $\mathrm{d}(1, 2) \leq 5$, hence $\mathrm{d}(n, 2n) \leq 5$. For the lower bound, the radius-2 balls are $B_2(1) = \{1, 4, 16, 18, 25\}$ and $B_2(2) = \{2, 8, 9, 32, 36, 48, 50, 81, 100\}$; they are disjoint, so no path of length ≤ 4 joins 1 and 2. Thus $\mathrm{d}(1, 2) = 5$. □

3 Selector models

Definition 3.1 (Selector)**.** A *selector* is a pair (Q, Σ) of a finite set Q with a distinguished element READY and a function $\Sigma : \mathbb{N}_{>0} \times Q \to \{+, -\} \times \mathbb{N}_{>0} \times Q$. A configuration (x, q) evolves by $(x, q) \mapsto (x', q')$ where $\Sigma(x, q) = (\sigma, N, q')$ and x' is the result of the move σN from x; Σ must return only legal moves. The only *admissible computation* inside Σ is exact arithmetic on the current integer x (addition, multiplication, exact division, comparison, integer roots, and inversion of strictly increasing integer polynomials); Σ receives no target sequence, counter or other external datum.

Definition 3.2 (Realization)**.** A trajectory *realizes* $(a_k)_{k \geq 1}$ *as raw waypoints* if $a_1, a_2, \ldots$ occur among the integers $x_0, x_1, \ldots$ at strictly increasing times $\tau_1 < \tau_2 < \cdots$. It realizes (a_k) *at completion times* if in addition one may take τ_k to be the successive visits of the control state READY.

Four classes are compared.

class	definition	extra input
$\mathcal{S}_0$ state-only	$\lvert Q\rvert = 1$ (a function of x)	none
$\mathcal{S}_1$ finite control	$\lvert Q\rvert < \infty$, arithmetic on x as above	none
$\mathcal{S}_2$ unbounded memory	Q replaced by a counter or tape	none
$\mathcal{S}_3$ external targets	Σ may read the target sequence	the sequence

Here $\mathcal{S}_0 \subseteq \mathcal{S}_1 \subseteq \mathcal{S}_2 \subseteq \mathcal{S}_3$ by definition. Which inclusions are strict for the identity sequence is the content of Sections 4–6.

4 The memoryless obstruction

Theorem 4.1 (U2.0)**.** *No $\mathcal{S}_0$ selector, i.e. no function F with $F(x)$ a neighbour of x, has an orbit containing $1, 2, 3, \ldots$ as raw waypoints.*

Proof. Suppose the orbit (x_j) does. If some value is revisited, $x_i = x_j$ with $i < j$, determinism makes the orbit eventually periodic, so it takes only finitely many values and cannot contain all positive integers. Hence the orbit is injective, and each integer m occurs exactly once, at time τ_m, with $\tau_1 < \tau_2 < \cdots$. Every x_j is a positive integer m, so $j = \tau_m$. If $\tau_{m+1} > \tau_m + 1$, the state x_j at some j with $\tau_m < j < \tau_{m+1}$ would be an integer m' with $\tau_{m'} = j$, contradicting the monotonicity of τ; and the same argument excludes states before τ_1. So $\tau_{m+1} = \tau_m + 1$ and $x_{j+1} = x_j + 1$, i.e. $\{m, m+1\}$ is an edge, contradicting Lemma 2.1. □

Remark 4.2. Only Lemma 2.1 is used; the bound $\mathrm{d} \leq 6$ plays no role. The obstruction holds for any graph on $\mathbb{N}_{>0}$ in which no two consecutive integers are adjacent.

5 Finite-control successor generation

5.1 The selector

For each class $c \in \{A, \ldots, I\}$ let $n = Mt + r$ $(t \geq t_0)$ be its residue form, with moves $\sigma_1 N_1, \ldots, \sigma_\ell N_\ell$ whose labels $N_i = N_i(t)$ are given in Appendix A, and let $s_0(t) = n$, $s_1(t), \ldots, s_\ell(t) = n + 1$ be the successive states. The control set is

$$Q = \{\mathsf{READY}\} \cup \{(c, i) : c \in \{A, \ldots, I\},\, 1 \leq i < \ell_c\} \cup \{(S_1, i)\}_{i=1}^{4} \cup \{(S_3, 1)\},$$

of size 43. The selector is:

- $\Sigma(1, \mathsf{READY})$ and $\Sigma(3, \mathsf{READY})$ begin the fixed special paths $1 \to 4 \to 18 \to 81 \to 8 \to 2$ and $3 \to 16 \to 4$ (finitely many table entries).
- $\Sigma(x, \mathsf{READY})$ for other x: let c be the class of the residue of x (mod24), let t be the unique solution of $s_0(t) = x$, and play move $\sigma_1 N_1(t)$ with new control $(c, 1)$ (or READY if $\ell_c = 1$).
- $\Sigma(x, (c, i))$: let t be the unique solution of $s_i(t) = x$ with $t \geq t_0$, play move $\sigma_{i+1} N_{i+1}(t)$, and change control to $(c, i + 1)$, or to READY if $i + 1 = \ell_c$.

Lemma 5.1 (Recoverability)**.** *For each class c and each $0 \leq i \leq \ell_c$ the state $s_i(t)$ is a strictly increasing polynomial on $t \geq t_0$ with rational coefficients, taking integer values. Hence t is determined by (c, i, x) and is computed by exact arithmetic (for example binary search on a monotone integer polynomial).*

Proof. The states $s_i(t)$ are listed in Appendix A in factored form; each is a product of positive powers of linear forms in t with positive coefficients on $t \geq t_0$, hence strictly increasing. Symbolic verification: check W3 of Table 1. □

Remark 5.2 (Example)**.** In class A $(n = 4t + 1)$: $4t + 1 \to 4(2t + 1)^2 \to 8t \to 8(t + 1)^2 \to 2(2t + 1)$. At the third state $t = x/8$, at the fourth $t = \sqrt{x/8} - 1$.

Theorem 5.3 (U2.1)**.** *The selector Σ is a legal $\mathcal{S}_1$ selector. From $(1, \mathsf{READY})$ its trajectory returns to* READY *exactly at $2, 3, 4, \ldots$ in order; each return takes between 2 and 6 moves.*

Proof. By Lemma 5.1, Σ is a well-defined function of (x, q). For $x = n \geq 1$ in READY with $n \notin \{1, 3\}$, the parameter t is determined and the states visited are $s_1(t), \ldots, s_\ell(t) = n + 1$ because the identities of Lemma 2.2 are exactly these compositions; every move is legal by the certificates of [3]. At $n + 1$ control returns to READY. The special paths end at 2 and 4. Induction on n completes the proof. □

Remark 5.4 (Consistency with U2.0)**.** Projecting onto x alone, the same integer occurs under different controls with different moves (the verifier exhibits one), so the trajectory is not the orbit of any $\mathcal{S}_0$ selector. The control component is only a finite step counter, and the template index fixed by the residue of n; no counter grows with n.

Corollary 5.5 (Cost)**.** *After K waypoints, at most $6K$ moves have been made.*

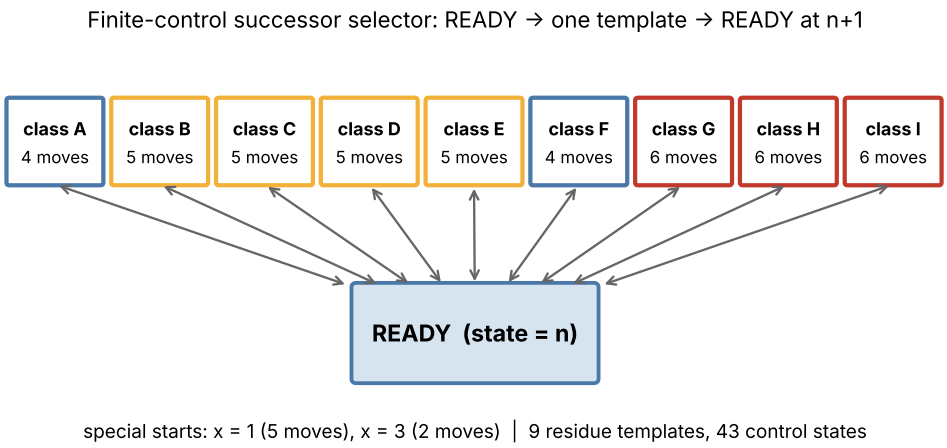


Figure 1: Control structure of Σ. From READY a residue class selects one of nine templates; the template runs to completion and returns to READY at $n + 1$.

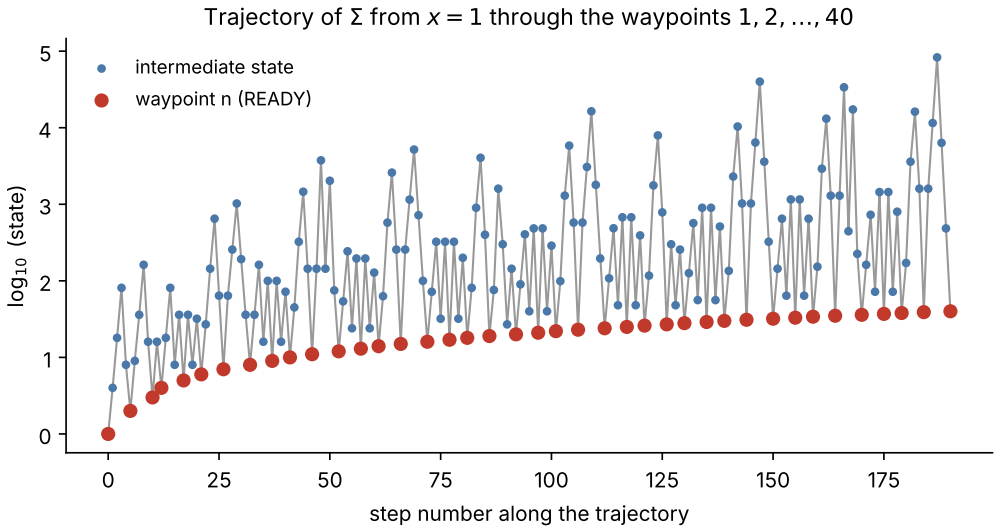


Figure 2: The trajectory of Σ from $x = 1$ through the first forty waypoints on a logarithmic scale.

5.2 Computational validation

The long-run test below is a validation of the formal argument, not a substitute for it. Over 100 000 successive waypoints the selector made 491 665 moves, mean 4.9166 per waypoint, maximum 6; every move was checked against the edge definition.

steps to next waypoint	count
2	1
4	29 165
5	50 001
6	20 833

6 What the classes can and cannot distinguish

Proposition 6.1 (Separation for the identity sequence)**.** *The identity sequence is realized as raw waypoints by $\mathcal{S}_1$ (Theorem 5.3) and not by $\mathcal{S}_0$ (Theorem 4.1). Therefore $\mathcal{S}_0 \subsetneq \mathcal{S}_1$ at the level of realizable sequences.*

Proposition 6.2 (Unrestricted observation trivializes the comparison)**.** *Let $F(x) = 4x$ (the move $+1$, legal at every x). Its orbit from $x = 1$ is 4^j. For any computable sequence (a_k) the observation $o(x) = a_{\log_4 x}$ (defined on powers of 4) turns this state-only selector into one whose observations realize (a_k).*

Consequently an observation map cannot be an arbitrary computable function. In this release a realization is always read directly from the integer states (raw waypoints) or from visits to READY; any further restriction on observation, such as bounded-memory or length-preserving read-outs, is an open design choice and is not needed for the results above.

7 A criterion for native generation

Definition 7.1 (Native realization)**.** A selector (Q, Σ) *natively realizes* (a_k) if (1) Σ is one fixed finite object, independent of k; (2) it receives no external input; (3) (a_k) occurs as raw waypoints or at READY completion times; (4) the only unbounded quantity it uses at any step is the current integer.

Theorem 5.3 says that the identity sequence is natively realized in $\mathcal{S}_1$. What was supplied is finite: nine templates and two special paths. What was not supplied is the sequence. In this sense the selector generates $1, 2, 3, \ldots$ and does not merely traverse it.

Limits of the claim. The result concerns a sequence whose consecutive terms are uniformly close in G; the selector encodes the one finite fact "how to reach $n + 1$ from n in at most six moves". The criterion does not yet say which other sequences admit such a selector.

8 Status and open questions

Open questions.

1. Which sequences other than $1, 2, 3, \ldots$ are natively realized in $\mathcal{S}_1$: squares, primes, Fibonacci numbers? For squares one needs a uniform bound on $\mathrm{d}(n^2, (n+1)^2)$ and a decodable family of path identities; for primes the unbounded search seems to need more than finite control.
2. Is every sequence with uniformly bounded waypoint distance and a recoverable path family realizable in $\mathcal{S}_1$? What replaces recoverability when the intermediate states are not monotone in a single parameter?
3. What is the right restriction on observation maps so that comparisons between selector classes remain informative?

statement	status	basis
U2.0 obstruction	proved	elementary argument; ingredient checked to $n \le 20000$ (W7)
Recoverability	proved	polynomial monotonicity; symbolic check, 55 states (W3)
U2.1 realization	proved, computer-supported	uses certified identities of [3]; long-run validation (W5, W6)
Doubling = 5	proved	Lemma 2.3; replay to 2×10^5 and two-engine ball check (W1, W2)
Formal proof-assistant certification	not done	–
Literature originality review	not done	–

Table 1: Status of the results. All verifier checks passed (323 103 cases; verifier hash prefix `26e8c9f0c3abecda`).

A The nine templates

Each line lists the states and the moves (labels over arrows) of one class, in factored form, from the symbolic composition of the certified identities.

Class A. $n = 4t + 1$, $t \ge 1$:

$$4t+1 \xrightarrow{+(4t+1)} 4\,(2t+1)^2 \xrightarrow{-2t} 8t$$
$$8t \xrightarrow{+t} 8\,(t+1)^2 \xrightarrow{-(2t+1)} 2\,(2t+1)$$

Class B. $n = 8t$, $t \ge 1$:

$$8t \xrightarrow{+2} 36t \xrightarrow{+2} 162t$$
$$162t \xrightarrow{-8} 16t \xrightarrow{+4t} 4\,(4t+1)^2$$
$$4\,(4t+1)^2 \xrightarrow{-(8t+1)} 8t+1$$

Class C. $n = 8t + 2$, $t \ge 0$:

$$2\,(4t+1) \xrightarrow{+2} 9\,(4t+1) \xrightarrow{+(4t+1)} 36\,(2t+1)^2$$
$$36\,(2t+1)^2 \xrightarrow{+2} 162\,(2t+1)^2 \xrightarrow{-8} 16\,(2t+1)^2$$
$$16\,(2t+1)^2 \xrightarrow{-(8t+3)} 8t+3$$

Class D. $n = 8t + 4$, $t \ge 0$:

$$4\,(2t+1) \xrightarrow{+2} 18\,(2t+1) \xrightarrow{+2} 81\,(2t+1)$$
$$81\,(2t+1) \xrightarrow{-8} 8\,(2t+1) \xrightarrow{+2(2t+1)} 4\,(4t+3)^2$$
$$4\,(4t+3)^2 \xrightarrow{-(8t+5)} 8t+5$$

Class E. $n = 8t + 6$, $t \ge 0$:

$$2\,(4t+3) \xrightarrow{+2} 9\,(4t+3) \xrightarrow{+(4t+3)} 144\,(t+1)^2$$
$$144\,(t+1)^2 \xrightarrow{+2} 648\,(t+1)^2 \xrightarrow{-8} 64\,(t+1)^2$$
$$64\,(t+1)^2 \xrightarrow{-(8t+7)} 8t+7$$

Class F. $n = 24t + 3$, $t \geq 1$:

$$3\,(8t+1) \xrightarrow{+(8t+1)} 12\,(4t+1)^2 \xrightarrow{-4t} 48t$$
$$48t \xrightarrow{+3t} 16\,(3t+1)^2 \xrightarrow{-(6t+1)} 4\,(6t+1)$$

Class G. $n = 8t + 7$, $t \geq 0$:

$$8t+7 \xrightarrow{+(8t+7)} 64\,(t+1)^2 \xrightarrow{+(t+1)} 64\,(t+1)\,(t+2)^2$$
$$64\,(t+1)\,(t+2)^2 \xrightarrow{+(t+1)} 64\,(t+2)^4 \xrightarrow{-(2t+3)} 16\,(t+2)^2\,(2t+3)$$
$$16\,(t+2)^2\,(2t+3) \xrightarrow{-(2t+3)} 4\,(2t+3)^2 \xrightarrow{-2(t+1)} 8\,(t+1)$$

Class H. $n = 24t + 11$, $t \geq 0$:

$$24t+11 \xrightarrow{+(24t+11)} 144\,(2t+1)^2 \xrightarrow{+24} 3750\,(2t+1)^2$$
$$3750\,(2t+1)^2 \xrightarrow{-2(25t+12)} 12\,(25t+12) \xrightarrow{+(25t+12)} 12\,(25t+13)^2$$
$$12\,(25t+13)^2 \xrightarrow{-25(2t+1)} 75\,(2t+1) \xrightarrow{-4} 12\,(2t+1)$$

Class I. $n = 24t + 19$, $t \geq 0$:

$$24t+19 \xrightarrow{+1} 4\,(24t+19) \xrightarrow{+(24t+19)} 64\,(6t+5)^2$$
$$64\,(6t+5)^2 \xrightarrow{-3} 12\,(6t+5)^2 \xrightarrow{-3(4t+3)} 9\,(4t+3)$$
$$9\,(4t+3) \xrightarrow{+(4t+3)} 144\,(t+1)^2 \xrightarrow{-(6t+5)} 4\,(6t+5)$$

B Reproducibility

From the directory `code/`: `python3 verify_u2_001.py --out ../reports/u2_001_verification_report.j --data ../data` (about three minutes). The files `u1_bnd_lib.py` (unchanged from the U1-BND release) and `u2_selector.py` define the graph primitives and Σ. Use `python3` without the `-I` flag, from `code/`. The verifier does not re-run the heavy checks of the earlier releases (recorded as NOT_RERUN with hashes in the README).

References

[1] UNNS Research Collective, *UNNS-U0: Arithmetic reachability and orbit admissibility*, research note U0-001, 8 November 2026 (local release; not part of the frozen repository).

[2] UNNS Research Collective, *UNNS-U1: Reversible arithmetic connectivity*, research note, 8 November 2026 (provisional baseline; local release).

[3] UNNS Research Collective, *Bounded Successor Distance in the Reversible Square-Multiplication Graph*, UNNS-U1-BND version 1.0 (v001), 8 November 2026 (local release).

[4] UNNS Research Collective, *UNNS Generative Information Geometry collection*, frozen release, repository `ukbbi/UNNS`, commit `a115e4bc6b8ee734e6146b3b17a6b121e65d31c3`: https://github.com/ukbbi/UNNS/tree/a115e4bc6b8ee734e6146b3b17a6b121e65d31c3/projects/UNNS_Generative_Information_Geometry.

[5] UNNS project repository, pinned at the same commit: https://github.com/ukbbi/UNNS/tree/a115e4bc6b8ee734e6146b3b17a6b121e65d31c3.